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6.3 Online Question Bank Instructions

Accessing the Data Insights Practice Questions

Below are Data Sufficiency and Two-Part Analysis Data Insights practice questions. Because the Data Insights questions need to be rendered online, certain question types, such as Multi-Source Reasoning, Table Analysis, and Graphics Interpretation, are only available in the **Online Question Bank** that is included as a part of this book.

Use your unique Access Code from the inside front cover of this book to access additional practice questions with answer explanations at **www.mba.com/my-account**. You can buy more Data Insights practice questions on **www.mba.com/gmatprep**.

To register for the GMAT™ exam, go to www.mba.com/register

6.4 Practice Questions: Data Sufficiency

Each Data Sufficiency question has a math problem and two statements, labeled (1) and (2), which present data. Using this data with your knowledge of math and everyday facts (such as the number of days in July or what *counterclockwise* means), decide whether the data in the statements are enough to solve the problem. Then pick one of these answer choices:

A. **Statement (1) ALONE is sufficient, but statement (2) alone is not sufficient.**

B. **Statement (2) ALONE is sufficient, but statement (1) alone is not sufficient.**

C. **BOTH statements TOGETHER are sufficient, but NEITHER statement ALONE is sufficient.**

D. **EACH statement ALONE is sufficient.**

E. **Statements (1) and (2) TOGETHER are NOT sufficient.**

Note: In Data Sufficiency questions that ask for a quantity's value, the data given in the statements are sufficient only when they make it possible to find exactly one numerical value for the quantity.

Questions 273 to 319 — Difficulty: Easy

273. What is the number of pages of a certain journal article?

1. The size of each page is $5\frac{1}{2}$ inches by 8 inches.
2. The average (arithmetic mean) number of words per page is 250.

274. If a certain vase contains only roses and tulips, how many tulips are there in the vase?

1. The number of roses in the vase is 4 times the number of tulips in the vase.
2. There is a total of 20 flowers in the vase.

275. The cost of 10 pounds of apples and 2 pounds of grapes was $12. What was the cost per pound of apples?

1. The cost per pound of grapes was $2.
2. The cost of 2 pounds of apples was less than the cost of 1 pound of grapes.

276. What was the median annual salary for the employees at Company X last year?

1. Last year, there were 29 employees at Company X.
2. Last year, 12 employees at Company X had an annual salary of $24,000.

277. How many basic units of Currency X are equivalent to 250 basic units of Currency Y?

1. 100 basic units of Currency X are equivalent to 625 basic units of Currency Y.
2. 2,000 basic units of Currency X are equivalent to 12,500 basic units of Currency Y.

278. A company bought 3 printers and 1 scanner. What was the price of the scanner?

1. The total price of the printers and the scanner was $1,300.
2. The price of each printer was 4 times the price of the scanner.

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2. The price of each printer was 4 times the price of the scanner.

279. Each of the 256 solid-colored marbles in a box is either blue, green, or purple. What is the ratio of the number of blue marbles to the number of purple marbles in the box?

1. The number of green marbles in the box is 4 times the number of blue marbles in the box.
2. There are 192 green marbles in the box.

280. A certain mixture of paint requires blue, yellow, and red paints in ratios of 2:3:1, respectively, and no other ingredients. If there are ample quantities of the blue and red paints available, is there enough of the yellow paint available to make the desired amount of the mixture?

1. Exactly 20 quarts of the mixture are needed.
2. Exactly 10 quarts of the yellow paint are available.

281. There are 2 groups of students who took a history test. Was the average (arithmetic mean) score of the students in Group A higher than the average score of the students in Group B who took the test?

1. Of the students who took the test, 10 were in Group A and 12 were in Group B.
2. On the test, the highest score achieved was achieved by a Group B student, and the lowest score achieved was achieved by a Group A student.

282. The research funds of a certain company were divided among three departments, X, Y, and Z. Which one of the three departments received the greatest proportion of the research funds?

1. The research funds received by Departments X and Y were in the ratio 3 to 5, respectively.
2. The research funds received by Departments X and Z were in the ratio 2 to 1, respectively.

283. In a certain class, some students donated cans of food to a local food bank. What was the average (arithmetic mean) number of cans donated per student in the class?

1. The students donated a total of 56 cans of food.
2. The total number of cans donated was 40 greater than the total number of students in the class.

284. Each of the n employees at a certain company has a different annual salary. What is the median of the annual salaries of the n employees?

1. When the annual salaries of the n employees are listed in increasing order, the median is the 15th salary.
2. The sum of the annual salaries of the n employees is $913,500.

285. In a recent town election, what was the ratio of the number of votes in favor of a certain proposal to the number of votes against the proposal?

1. There were 60 more votes in favor of the proposal than against the proposal.
2. There were 240 votes in favor of the proposal.

286. How many men are in a certain company's vanpool program?

1. The ratio of men to women in the program is 3 to 2.
2. The men and women in the program fill 6 vans.

287. Last semester, Professor K taught two classes, A and B. Each student in Class A handed in 7 assignments, and each student in Class B handed in 5 assignments. How many students were in Class A?

1. The students in both classes combined handed in a total of 85 assignments.
2. There were 10 students in Class B.

288. Was the amount of John's heating bill for February greater than it was for January?

1. The ratio of the amount of John's heating bill for February to that for January was $\frac{26}{25}$.
2. The sum of the amounts of John's heating bills for January and February was $183.60.

289. Machine R and Machine S work at their respective constant rates. How much time does it take Machine R, working alone, to complete a certain job?

1. [illegible] $\frac{3}{}$ [illegible]

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1. The amount of time that it takes Machine S, working alone, to complete the job is $\frac{3}{4}$ the amount of time that it takes Machine R, working alone, to complete the job.
2. Machine R and Machine S, working together, take 12 minutes to complete the job.

290. Beth's bank charges a service fee on a regular checking account for each month in which the balance on the account falls below $100 at any time during the month. Did the bank charge a service fee on Beth's regular checking account last month?
    1. During last month, a total of $1,000 was withdrawn from Beth's regular checking account.
    2. At the beginning of last month, Beth's regular checking account balance was $500.

291. Three houses are being sold through a real estate agent. What is the asking price for the house with the second-largest asking price?
    1. The difference between the greatest and the least asking price is $130,000.
    2. The difference between the two greater asking prices is $85,000.

292. How many people in a group of 50 own neither a fax machine nor a laser printer?
    1. The total number of people in the group who own a fax machine or a laser printer or both is less than 50.
    2. The total number of people in the group who own both a fax machine and a laser printer is 15.

293. A state legislature had a total of 96 members. The members who did not vote on a certain bill consisted of 25 who were absent and 3 who abstained. How many of those voting voted for the bill?
    1. Exactly $\frac{1}{3}$ of the total membership of the legislature voted against the bill.
    2. The number of legislators who voted for the bill was 8 more than the total number who were absent or abstained.

294. What is the average (arithmetic mean) of the first 90 measurements in a certain list of 92 measurements?
    1. The average of the 92 measurements is 7.0 centimeters.
    2. The average of the last 2 measurements is 7.2 centimeters.

295. How many of the employees of a certain company use both a tablet and an iPhone?
    1. 60 of the employees use a tablet.
    2. 45 of the employees use an iPhone.

296. What is the standard deviation of the weights of the 30 samples in a certain experiment?
    1. The total weight of the 30 samples is 360 grams.
    2. Each of the 30 samples weighs 12 grams.

297. The people in a line waiting to buy tickets to a show are standing one behind the other. Adam and Beth are among the people in the line, and Beth is standing behind Adam with a number of people between them. If the number of people in front of Adam plus the number of people behind Beth is 18, how many people in the line are behind Beth?
    1. There are a total of 32 people in the line.
    2. 23 people in the line are behind Adam.

298. In a certain group of 50 people, how many are doctors who have a law degree?
    1. In the group, 36 people are doctors.
    2. In the group, 18 people have a law degree.

299. Of a group of 50 households, how many have at least one cat or at least one dog, but not both?
    1. The number of households that have at least one cat and at least one dog is 4.
    2. The number of households that have no cats and no dogs is 14.

300. Robin invested a total of $12,000 in two investments, X and Y, so that the investments earned the same amount of simple annual interest. How many dollars did Robin invest in Investment X?

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of simple annual interest. How many dollars did Robin invest in Investment Y?

1. Investment X paid 3 percent simple annual interest, and Investment Y paid 6 percent simple annual interest.
2. Robin invested more than $1,000 in Investment X.

301. In a real estate office that employs n salespeople, f of them are females and x of the females are new employees. What is the value of n?

1. If an employee were randomly selected from the n employees, the probability of selecting a female would be $\frac{2}{3}$.
2. If an employee were randomly selected from the f female employees, the probability of selecting a new employee would be $\frac{1}{2}$.

302. Is the total price of the items in Andrea's shopping cart less than $25.00?

1. There are 5 items in Andrea's shopping cart.
2. The price of the most expensive item in Andrea's shopping cart is $4.99.

303. A certain plumber charges $92 for each job completed in 4 hours or less and $23 per hour for each job completed in more than 4 hours. If it took the plumber a total of 7 hours to complete two separate jobs, what was the total amount charged by the plumber for the two jobs?

1. The plumber charged $92 for one of the two jobs.
2. The plumber charged $138 for one of the two jobs.

304. If x and y are positive numbers, is $\frac{x+1}{y+1} > \frac{x}{y}$?

1. $x > 1$
2. $x < y$

305. If a school district paid a total of $35 per desk for x desks and a total of $30 per table for y tables, what was the total amount that the district paid for these desks and tables?

1. The total amount the district paid for the y tables was $900.
2. $x = 90$, and the total amount the district paid for the x desks was 3.5 times the total amount the district paid for the y tables.

306. Three children inherited a total of X dollars. If the oldest child inherited $7,000 more than the youngest child, and the youngest child inherited $9,000 less than the middle child, what is the value of X?

1. The middle child inherited $27,000.
2. The youngest child and the middle child together inherited a total of $45,000.

307. If a building has 6,000 square meters of floor space, how many offices are in the building?

1. Exactly $\frac{1}{4}$ of the floor space is not used for offices.
2. There are exactly 20 executive offices, and each of these occupies 3 times as much floor space as the average for all of the remaining offices.

308. For which type of investment, J or K, is the annual rate of return greater?

1. Type J returns $115 per $1,000 invested for any one-year period, and Type K returns $300 per $2,500 invested for any one-year period.
2. The annual rate of return for an investment of Type K is 12%.

309. If Car X followed Car Y across a certain bridge that is $\frac{1}{2}$ mile long, how many seconds did it take Car X to travel across the bridge?

1. Car X drove onto the bridge exactly 3 seconds after Car Y drove onto the bridge and drove off the bridge

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1. Car X drove onto the bridge exactly 3 seconds after Car Y drove onto the bridge and drove off the bridge exactly 2 seconds after Car Y drove off the bridge.
2. Car Y traveled across the bridge at a constant speed of 30 miles per hour.

310. Will Candidate A win the most votes of any of the competing candidates in the upcoming election?

1. An opinion poll, Poll P, indicates that 60% of respondents prefer Candidate A over other candidates competing in the upcoming election.
2. An election-forecasting opinion poll is a method to estimate election outcomes based on a sample that is representative of the population of all of those legally entitled to vote.

311. The government of Nation X makes policy decisions intended to advance the public interest, such as policy decisions aiming to promote employment, economic flourishing, and environmental protection. A policy decision can involve a significant trade-off, i.e., it can entail foreseeably giving up part or all of one valued outcome in the pursuit of another valued outcome. Does a policy decision by Nation X to permit extensive clearcutting of forests provide a valued outcome at the expense of another valued outcome?

1. Extensive clearcutting of forests is permitted in Nation X because it provides immediate economic gains by enabling businesses such as farming and lumbering to flourish and provide employment.
2. Extensive clearcutting of forests is permitted in Nation X even though it reduces forests' absorption of carbon dioxide, and this indirectly contributes to global warming, resulting in adverse climate changes that are certain to be very costly for Nation X to manage.

312. Is Sarah's mathematics test score higher than the class average?

1. Sarah scored 90% on her mathematics test, answering 45 out of 50 questions correctly.
2. Sarah's high percentage score indicates that she performed exceptionally well, with only a few questions answered incorrectly.

313. On Day X, was the forecast temperature 36°C at noon?

1. On Day X, the forecast temperature was 30°C at 9:00 a.m. and 40°C at 2:00 p.m.
2. On Day X, every half hour, the forecast temperature increased by 1°C.

314. Benjamin is a salesman working for commission at a shoe store. For the first $1,000 worth of sales in a given month, he earns a commission of 2%. For any sales he makes after the first $1,000 in that month, he earns a commission of 4%. What is the total dollar value of sales made by Benjamin in the month of March?

1. Benjamin earned a commission of $120 in March.
2. Benjamin doubled the dollar value of his sales in March compared to his sales in February.

315. What is the length of Train A if it crosses Bridge X in 30 seconds from the moment it enters the bridge to the moment its final carriage exits the bridge?

1. Train B, which is 200 meters long, traveling at a constant speed of 20 meters per second, crosses Bridge X in 40 seconds.
2. The end of Train A, which is traveling at a constant speed, passes a pole in 10 seconds from the moment the train meets the pole.

316. Solomon leases space for his barber shop, and his landlord has recently increased the rent by 40%. Solomon wants to increase his monthly revenue to pay the increased rent without affecting his net income. By what percentage must he increase his revenue?

1. 20% of his previous monthly revenue was used to pay the rent before the landlord increased it.
2. The new rent is $50 more than the previous rent.

317. What is the ratio of the amount, by weight, of Metal A to Metal B in Alloy K?

1. The ratio of Metal A to Metal B in Alloy K becomes 5:6 when 40 pounds of Metal A is added to the alloy.
2. The ratio of Metal A to Metal B in Alloy K becomes 7:6 when 60 pounds of Metal A is added to the alloy.

318. Train A started from Station X toward Station Y traveling at a constant speed, and on the same day, on a parallel track, Train B started from Station Y toward Station X at a constant speed. At what time did Train A reach Station Y?

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1. The two trains met at 3:00 p.m. The distance traveled by Train A was twice the distance traveled by Train B by the time they met.
2. The ratio between the speeds of Trains A and B is 3:5. Train B started from Station Y at 6:00 a.m.

319. Pipes A and B are the pipes that fill a tank. Pipes C and D are the pipes that empty the same tank. What is the capacity of the tank?
    1. Pipe A is twice as efficient as Pipe C, that is, the volume of water per hour that Pipe A moves is twice the volume per hour that Pipe C moves. Pipe A and Pipe B are equally efficient. When Pipes C and D are closed, Pipes A and B together fill the empty tank in 5 hours. Pipe C empties 6,000 liters per hour.
    2. Pipe B is three times as efficient as Pipe D.

Questions 320 to 362 — Difficulty: Medium

320. What is the total number of executives at Company P?
    1. The number of male executives is $\frac{3}{5}$ the number of female executives.
    2. There are 4 more female executives than male executives.

321. Jack picked 76 apples. Of these, he sold $4y$ apples to Juanita and $3t$ apples to Sylvia. If he kept the remaining apples, how many apples did he keep? (t and y are positive integers.)
    1. $y \geq 15$ and $t = 2$
    2. $y = 17$

322. The total price of 5 pounds of regular coffee and 3 pounds of decaffeinated coffee was $21.50. What was the price of the 5 pounds of regular coffee?
    1. If the price of the 5 pounds of regular coffee had been reduced 10 percent and the price of the 3 pounds of decaffeinated coffee had been reduced 20 percent, the total price would have been $18.45.
    2. The price of the 5 pounds of regular coffee was $3.50 more than the price of the 3 pounds of decaffeinated coffee.

323. A certain painting job requires a mixture of yellow, green, and white paint. If 12 quarts of paint are needed for the job, how many quarts of green paint are needed?
    1. The ratio of the amount of green paint to the amount of yellow and white paint combined needs to be 1 to 3.
    2. The ratio of the amount of yellow paint to the amount of green paint needs to be 3 to 2.

324. A company produces a certain toy in only 2 sizes, small or large, and in only 2 colors, red or green. If, for each size, there are equal numbers of red and green toys in a certain production lot, what fraction of the total number of green toys is large?
    1. In the production lot, 400 of the small toys are green.
    2. In the production lot, $\frac{2}{3}$ of the toys produced are small.

325. Jones has worked at Firm X twice as many years as Green, and Green has worked at Firm X four years longer than Smith. How many years has Green worked at Firm X?
    1. Jones has worked at Firm X 9 years longer than Smith.
    2. Green has worked at Firm X 5 years less than Jones.

326. On June 1, Mary paid Omar $360 for rent and utilities for the month of June. Mary moved out early, and Omar refunded the money she paid for utilities, but not for rent, for the days in June after she moved out. How many dollars did Omar refund to Mary?
    1. Mary moved out on June 24.
    2. The amount Mary paid for utilities was less than $\frac{1}{5}$ the amount Mary paid for rent.

327. A collection of 36 cards consists of 4 sets of 9 cards each. The 9 cards in each set are numbered 1 through 9. If one card has been removed from the collection, what is the number on that card?

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one card has been removed from the collection, what is the number on that card?

1. The units digit of the sum of the numbers on the remaining 35 cards is 6.
2. The sum of the numbers on the remaining 35 cards is 176.

328. What is the ratio of the number of cups of flour to the number of cups of sugar required in a certain cake recipe?

1. The number of cups of flour required in the recipe is 250 percent of the number of cups of sugar required in the recipe.
2. $1\frac{1}{2}$ more cups of flour than cups of sugar are required in the recipe.

329. A clothing manufacturer makes jackets that are wool or cotton or a combination of wool and cotton. The manufacturer has 3,000 pounds of wool and 2,000 pounds of cotton on hand. Is this enough wool and cotton to make at least 1,000 jackets?

1. Each wool jacket requires 4 pounds of wool, and no cotton.
2. Each cotton jacket requires 6 pounds of cotton, and no wool.

330. Are at least 10 percent of the people in Country X who are 65 years old or older employed?

1. In Country X, 11.3 percent of the population is 65 years old or older.
2. In Country X, of the population 65 years old or older, 20 percent of those with a college degree and 10 percent of those without a college degree are employed.

331. A scientist recorded the number of eggs in each of 10 birds' nests. What was the standard deviation of the numbers of eggs in the 10 nests?

1. The average (arithmetic mean) number of eggs for the 10 nests was 4.
2. Each of the 10 nests contained the same number of eggs.

332. A paint mixture was formed by mixing exactly 3 colors of paint. By volume, the mixture was $x\%$ blue paint, $y\%$ green paint, and $z\%$ red paint. If exactly 1 gallon of blue paint and 3 gallons of red paint were used, how many gallons of green paint were used?

1. $x = y$
2. $z = 60$

333. For a basic monthly fee of F yen ($¥F$), Naoko's first cell phone plan allowed him to use a maximum of 420 minutes on calls during the month. Then, for each of x additional minutes he used on calls, he was charged $¥M$, making his total charge for the month $¥T$, where $T = F + xM$. What is the value of F?

1. Naoko used 450 minutes on calls the first month, and the total charge for the month was ¥13,755.
2. Naoko used 400 minutes on calls the second month, and the total charge for the month was ¥13,125.

334. Is the sum of the prices of the 3 books that Shana bought less than $48?

1. The price of the most expensive of the 3 books that Shana bought is less than $17.
2. The price of the least expensive of the 3 books that Shana bought is exactly $3 less than the price of the second most expensive book.

335. If Elena spent a total of $720 on newspapers, magazines, and books last year, what amount did she spend on newspapers?

1. Last year, the amount that Elena spent on magazines was 80 percent of the amount that she spent on books.
2. Last year, the amount that Elena spent on newspapers was 60 percent of the total amount that she spent on magazines and books.

336. Of the 800 sweaters at a certain store, 150 are red. How many of the red sweaters at the store are made of pure wool?

1. 320 of the sweaters at the store are neither red nor made of pure wool.
2. 100 of the red sweaters at the store are not made of pure wool.

337. A certain computer company produces two different monitors, P and Q. In 2010, what was the net profit from

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337. A certain computer company produces two different monitors, P and Q. In 2010, what was the net profit from the sale of the two monitors?

    1. Of the company's expenses in 2010, rent and utilities totaled $500,000.
    2. In 2010, the company sold 50,000 units of Monitor P at $300 per unit and 30,000 units of Monitor Q at $650 per unit.

338. A conveyor belt moves bottles at a constant speed of 120 centimeters per second. If the conveyor belt moves a bottle from a loading dock to an unloading dock, is the distance that the conveyor belt moves the bottle less than 90 meters? (1 meter = 100 centimeters)

    1. It takes the conveyor belt less than 1.2 minutes to move the bottle from the loading dock to the unloading dock.
    2. It takes the conveyor belt more than 1.1 minutes to move the bottle from the loading dock to the unloading dock.

339. In a product test of a common cold remedy, x percent of the patients tested experienced side effects from the use of the drug and y percent experienced relief of cold symptoms. What percent of the patients tested experienced both side effects and relief of cold symptoms?

    1. Of the 1,000 patients tested, 15 percent experienced neither side effects nor relief of cold symptoms.
    2. Of the patients tested, 30 percent experienced relief of cold symptoms without side effects.

340. Three roommates—Bela, Gyorgy, and Janos—together saved money for a trip. The amount that Bela saved was equal to 8% of his monthly income. The amount that Gyorgy saved was exactly $\frac{1}{3}$ of the total amount saved by all 3 roommates. What was the total amount saved for the trip by all 3 roommates?

    1. Bela had a monthly income of $2,000.
    2. Janos saved 1.5 times as much for the trip as Bela.

341. In each game of a certain tournament, a contestant either loses 3 points or gains 2 points. If Pat had 100 points at the beginning of the tournament, how many games did Pat play in the tournament?

    1. At the end of the tournament, Pat had 104 points.
    2. Pat played fewer than 10 games.

342. At the beginning of the year, the Finance Committee and the Planning Committee of a certain company each had n members, and no one was a member of both committees. At the end of the year, 5 members left the Finance Committee and 3 members left the Planning Committee. How many members did the Finance Committee have at the beginning of the year?

    1. The ratio of the total number of members who left at the end of the year to the total number of members at the beginning of the year was 1:6.
    2. At the end of the year, 21 members remained on the Planning Committee.

343. Some computers at a certain company are Brand X and the rest are Brand Y. If the ratio of the number of Brand Y computers to the number of Brand X computers at the company is 5 to 6, how many of the computers are Brand Y?

    1. There are 80 more Brand X computers than Brand Y computers at the company.
    2. There is a total of 880 computers at the company.

344. Was the number of books sold at Bookstore X last week greater than the number of books sold at Bookstore Y last week?

    1. Last week, more than 1,000 books were sold at Bookstore X on Saturday and fewer than 1,000 books were sold at Bookstore Y on Saturday.
    2. Last week, less than 20 percent of the books sold at Bookstore X were sold on Saturday and more than 20 percent of the books sold at Bookstore Y were sold on Saturday.

345. From May 1 to May 30 in the same year, the balance in a checking account increased. What was the balance in the checking account on May 30?

    1. If, during this period of time, the increase in the balance in the checking account had been 12 percent, then

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